Advanced Git and Collaboration .

▢ Create a aws Account:

Create a new branch named "feature-aws-setup":

Navigate to Your Project Directory: Use a terminal or command prompt to navigate to your project directory.

Make Sure You Are Up-to-Date:

Update your main branch to ensure you have the latest changes.

git checkout main.

```
PS C:\Users\leela\Motion-git-assignment-2> git checkout main
A       aws-setup.md
Already on 'main'
```

git pull origin main.

Create the New Branch:

Create a new branch named "feature-aws-setup".

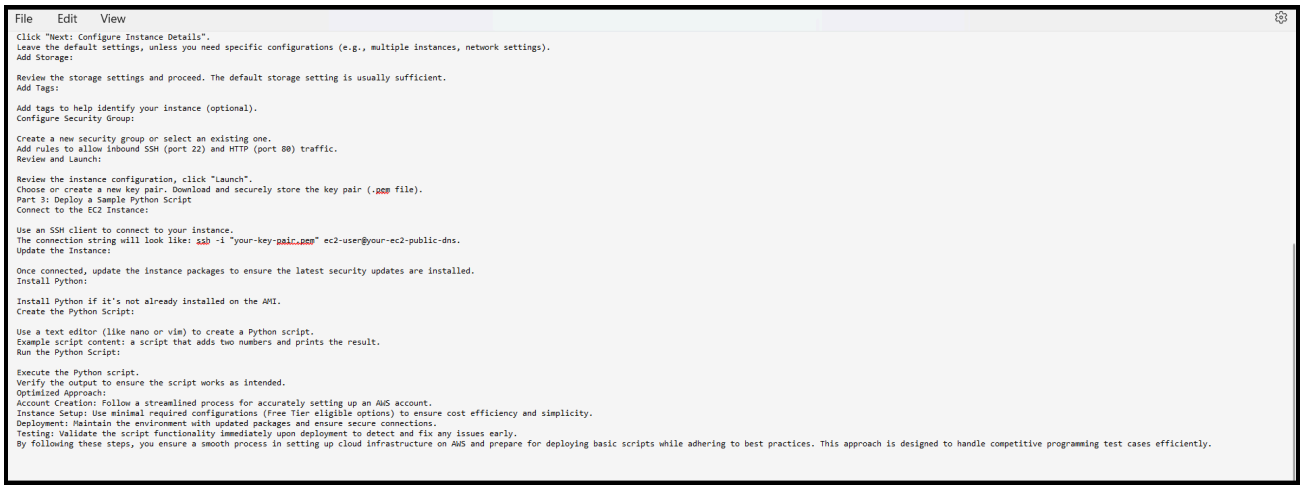
git checkout -b feature-aws-setup

```
PS C:\Users\leela\Motion-git-assignment-2> git checkout -b feature-aws-setup
Switched to a new branch 'feature-aws-setup'
```

b. Add a file named "aws-setup.md" to this branch:

```
PS C:\Users\leela\Motion-git-assignment-2> notepad aws-setup.md
```

Create the New File: Create a file named aws-setup.md in your project directory. You can use a text editor or run a touch command:



Document the Steps Required to Create an AWS Account, Set Up a Basic EC2 Instance, and Deploy a Sample Python Script:

Open aws-setup.md in your preferred text editor and document the following steps:

- **Steps to Create an AWS Account:**

1. Visit the AWS Website.

2. Sign Up for AWS.

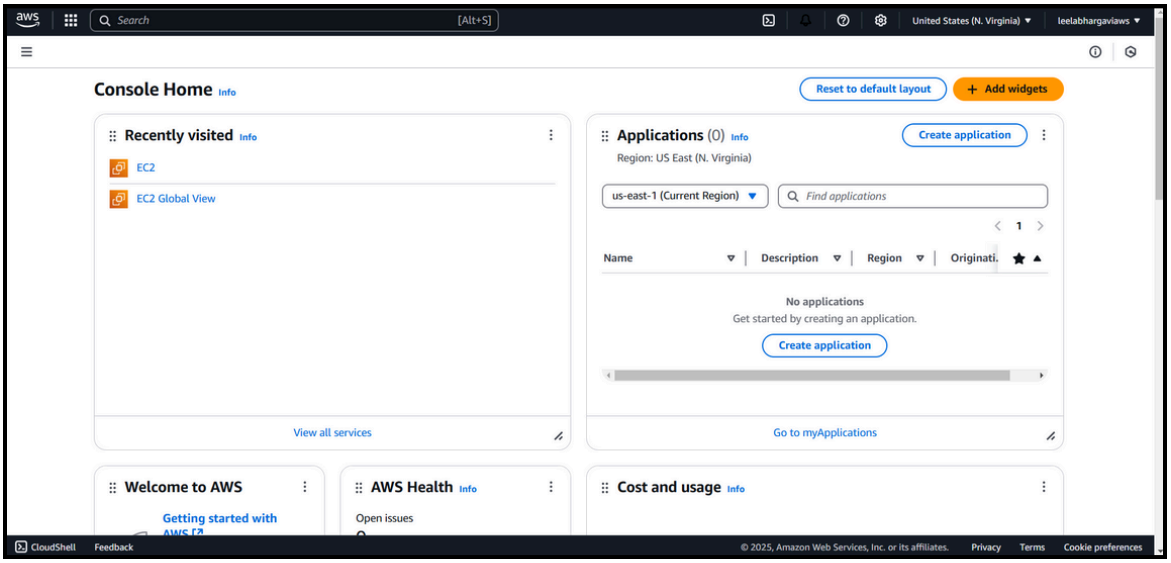
3. Verify Your Account.

4. Select an AWS Support Plan.

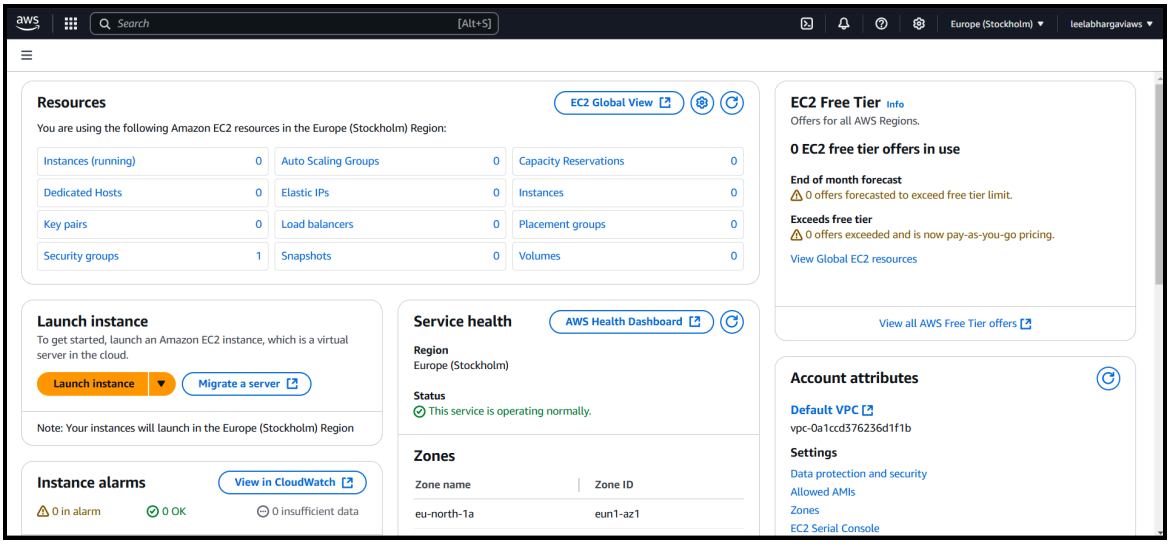
- **Steps to Set Up a Basic EC2 Instance:**

1. Log In to the AWS Management Console.

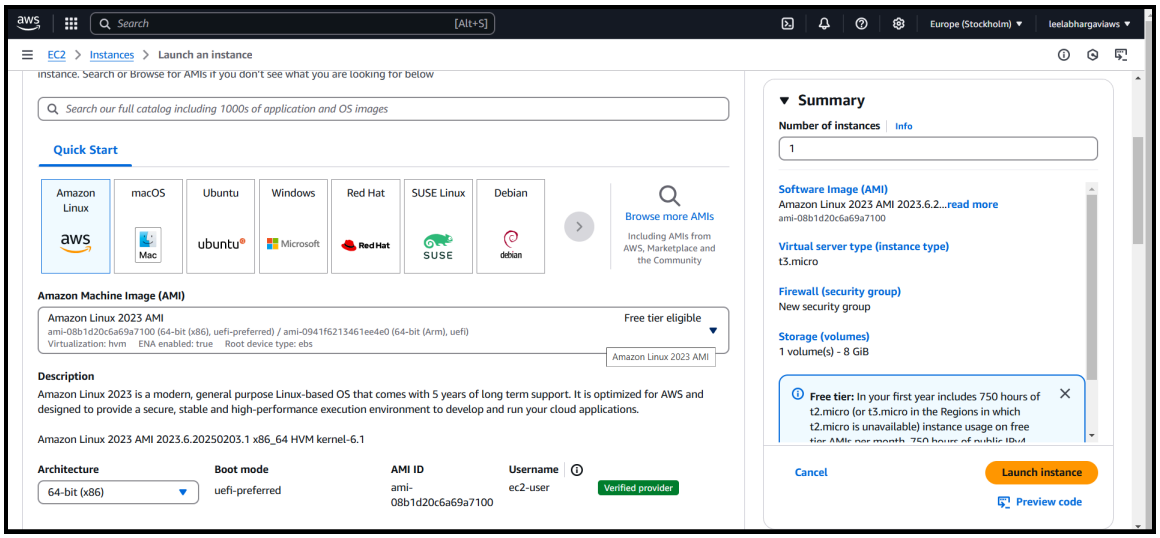
2. Navigate to EC2 Dashboard.



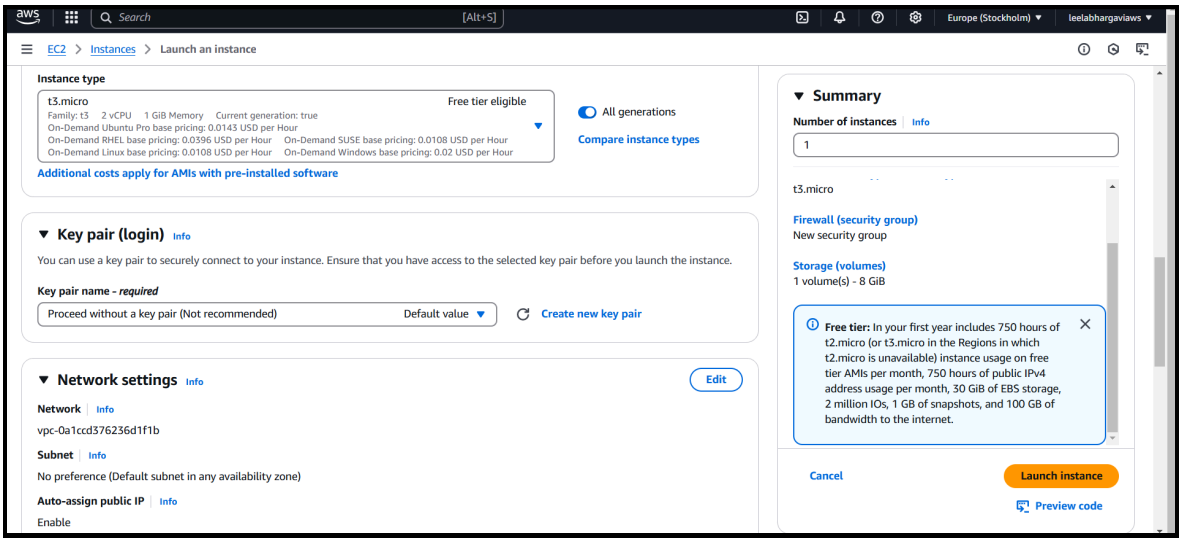
3. Launch an Instance.



4. Choose an Amazon Machine Image (AMI).



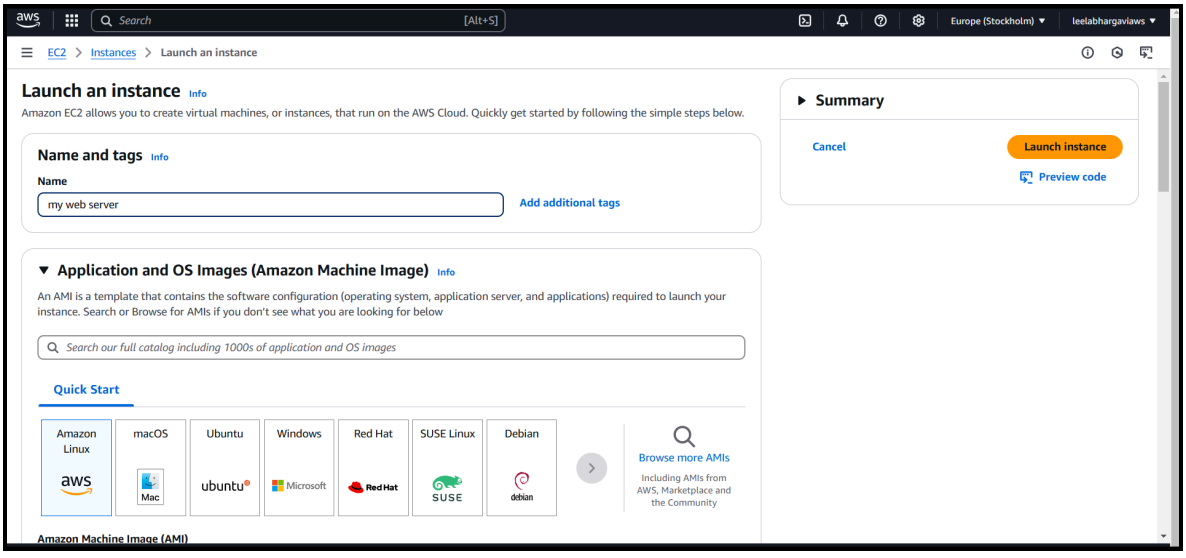
5. Choose an Instance Type.



6. Configure Instance Details.

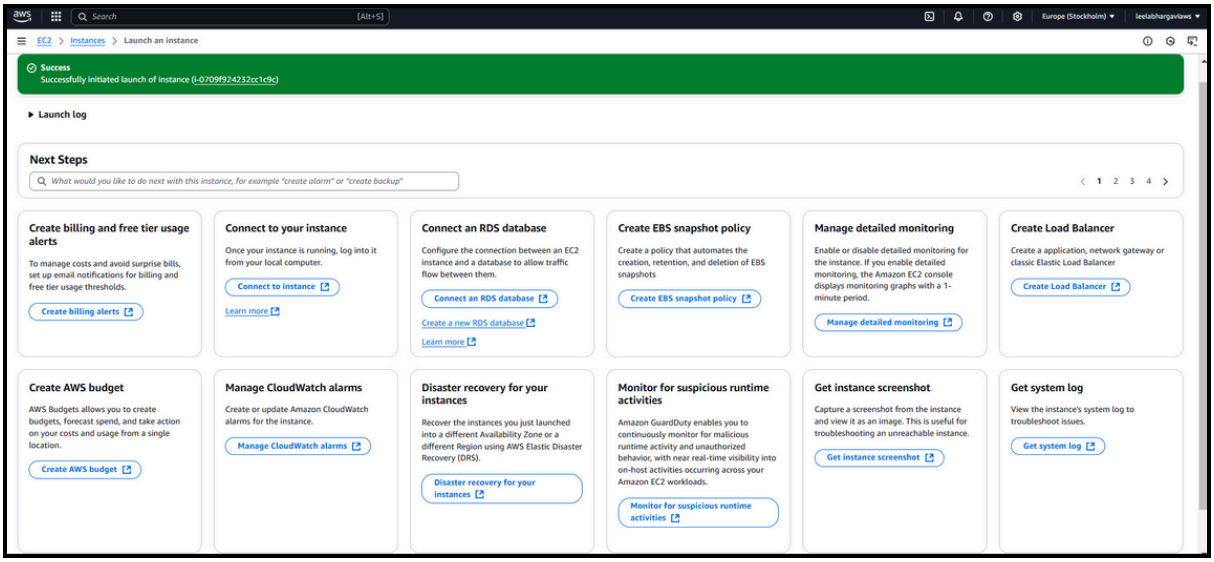
7. Add Name.

8. Add Tags.

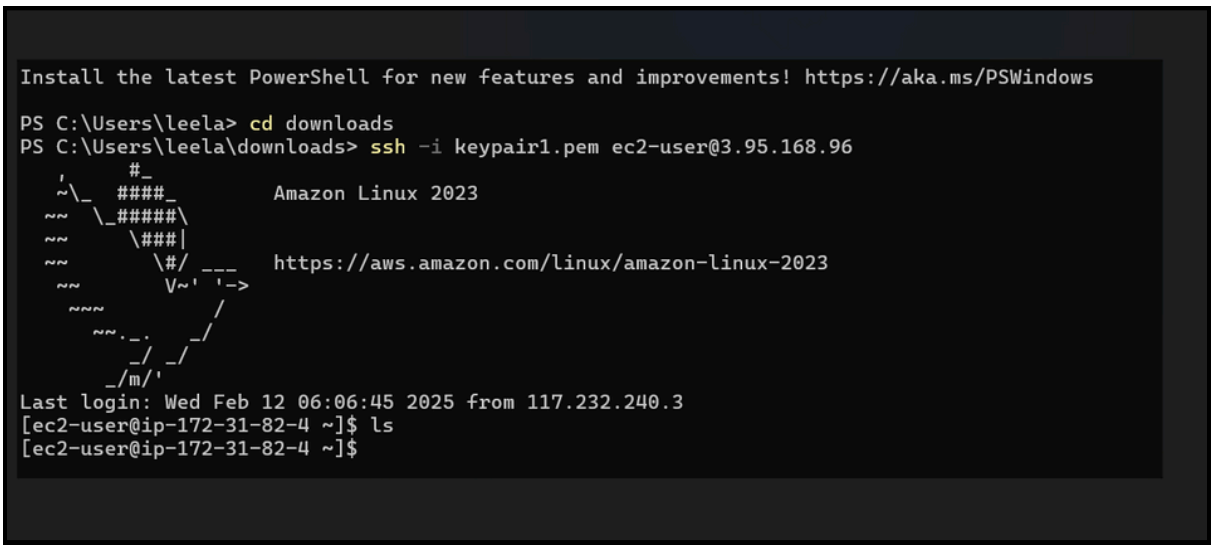


9. Configure Security Group.

10. Review and Launch.



11.now connect to git command ssh.



- **Steps to Deploy a Sample Python Script:**

1. Connect to Your EC2 Instance.

2. Install Required Packages.

3. Create the Python Script.

4. Run the Script.

- **Branching and Merging**

a. Create a new branch named “feature-aws-setup.”

b. Add a file named “aws-setup.md” to this branch. Inside the file, document the steps required to create an AWS account, set up a basic EC2 instance, and deploy a sample Python script (e.g., a script that adds two numbers).

c. Commit your changes with an appropriate message.

d. Merge the “feature-aws-setup” branch back into the main branch.

```
PS C:\Users\leela\motion-recruiment> git checkout -b feature-aws-setup
Switched to a new branch 'feature-aws-setup'
PS C:\Users\leela\motion-recruiment> notepad aws-setup.md
PS C:\Users\leela\motion-recruiment> git add aws-setup.md
PS C:\Users\leela\motion-recruiment> git commit -m "aws-setup"
[feature-aws-setup 1b7ad47] aws-setup
 1 file changed, 70 insertions(+)
 create mode 100644 aws-setup.md
PS C:\Users\leela\motion-recruiment> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\leela\motion-recruiment> git merge feature-aws-setup
Updating 6e12f3a..1b7ad47
Fast-forward
 aws-setup.md | 70 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 70 insertions(+)
 create mode 100644 aws-setup.md
PS C:\Users\leela\motion-recruiment> git push origin main
Enumerating objects: 4, done.
```

▢ Create an Azure Account:

- Go to website and create an azure free account.

2. Creating a machine in azure account.

- Navigate to Azure portal
- Now open Virtual Machines
- Choose the following instructions

    1. Enter a name for your virtual machine

    2. Choose image as Ubuntu

    3. And Size as Standard_B1s

    4. Now Create a authentication key

3. Deploy a python script.

4. Now create branch for Azure setup.

5. Commit it into the branch.

6. Push it into main branch.

- **Collaborative Development with GitHub:**

a. Create a new branch named “feature-azure-setup.”

b. Add a file named “azure-setup.md” to this branch. Inside the file, document the steps required to create an Azure account, set up a basic Azure VM, and deploy a sample Python script (e.g., a script that adds two numbers).

c. Commit your changes with a descriptive message.

d. Push your changes to the remote repository and create a pull request to merge “feature-azure-setup” into the main branch.

```
PS C:\Users\leela\motion-recruiment> git checkout -b feature-azure-setup
fatal: a branch named 'feature-azure-setup' already exists
PS C:\Users\leela\motion-recruiment> notepad azure-setup.md
PS C:\Users\leela\motion-recruiment> git add azure-setup.md
PS C:\Users\leela\motion-recruiment> git commit -m "azure-setup"
[main 1042f57] azure-setup
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\leela\motion-recruiment> git checkout main
Already on 'main'
Your branch is ahead of 'origin/main' by 3 commits.
  (use "git push" to publish your local commits)
PS C:\Users\leela\motion-recruiment> git merge feature-azure-setup
Already up to date.
PS C:\Users\leela\motion-recruiment> git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 3.16 KiB | 1.58 MiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/leelabhargavi29/motion-recruiment
   1b7ad47..1042f57  main -> main
```

▢ Create a GCP account:

- Go to website and create an gcp free account.

2. Creating a instances.

- Navigate to Compute Engine
- Open VM instances
- Click on create instance
- Choose the following instructions to create

    1. Choose Machine Type as e2-micro

    2. Now select OS as Debian-based

3. Deploy a python script.

4. Now create branch for Azure setup.

5. Commit it into the branch.

6. Push it into main branch.

- **GCP Deployment:**

a. Create a new branch named “feature-gcp-setup.”

b. Add a file named “gcp-setup.md” to this branch. Inside the file, document the steps required to create a GCP account, set up a Google Compute Engine instance, and deploy a sample Python script (e.g., a script that adds two numbers).

c. Commit your changes with a clear message.

d. Push your changes to the remote repository and create a pull request to merge “feature-gcp-setup” into the main branch.

e. Add a collaborator to review your pull request, address any feedback, and merge the changes once approved.

```
PS C:\Users\leela\motion-recruiment> git checkout -b feature-gcp-setup
Switched to a new branch 'feature-gcp-setup'
PS C:\Users\leela\motion-recruiment> notepad gcp-setup.md
PS C:\Users\leela\motion-recruiment> git add gcp-setup.md
PS C:\Users\leela\motion-recruiment> git commit -m "gcp-setup"
[feature-gcp-setup 13e0472] gcp-setup
 1 file changed, 44 insertions(+)
 create mode 100644 gcp-setup.md
PS C:\Users\leela\motion-recruiment> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\leela\motion-recruiment> git merge feature-gcp-setup
Updating 1042f57..13e0472
Fast-forward
 gcp-setup.md | 44 ++++++++++++++++++++++++++++++++++++++++++++
 1 file changed, 44 insertions(+)
 create mode 100644 gcp-setup.md
PS C:\Users\leela\motion-recruiment> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.29 KiB | 1.29 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/leelabhargavi29/motion-recruiment
   1042f57..13e0472  main -> main
```

- **Final Integration:**

a. After merging all three branches (AWS, Azure, and GCP), ensure that the main branch contains documentation for all three cloud setups in the corresponding markdown files.

b. Create a new branch named “feature-final-review.”

c. Review and finalize all documentation, ensuring clarity and completeness.

d. Merge the final review branch back into the main branch.

```
PS C:\Users\leela\motion-recruiment> git branch
  feature-aws-setup
  feature-azure-setup
  feature-gcp-setup
* main
PS C:\Users\leela\motion-recruiment> git checkout -b feature-final-review
Switched to a new branch 'feature-final-review'
PS C:\Users\leela\motion-recruiment> git checkout feature-final-review
Already on 'feature-final-review'
PS C:\Users\leela\motion-recruiment> git merge main
Already up to date.
PS C:\Users\leela\motion-recruiment> git push origin feature-final-review
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-final-review' on GitHub by visiting:
remote:      https://github.com/leelabhargavi29/motion-recruiment/pull/new/feature-final-review
remote:
To https://github.com/leelabhargavi29/motion-recruiment
 * [new branch]      feature-final-review -> feature-final-review
PS C:\Users\leela\motion-recruiment> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\leela\motion-recruiment> git commit -m "Assignment-3"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\leela\motion-recruiment> git branch -m master main
fatal: no branch named 'master'
PS C:\Users\leela\motion-recruiment> git branch -m feature-final-review
fatal: a branch named 'feature-final-review' already exists
PS C:\Users\leela\motion-recruiment> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\leela\motion-recruiment> git remote -v
origin  https://github.com/leelabhargavi29/motion-recruiment (fetch)
origin  https://github.com/leelabhargavi29/motion-recruiment (push)
PS C:\Users\leela\motion-recruiment> git push -u origin HEAD
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\leela\motion-recruiment>
```

